

3.4.3 Electric Fields

- **Coulomb's law**

Force between point charges in a vacuum

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2},$$

where ϵ_0 is the permittivity of free space.

- **Electric field strength**

E as force per unit charge defined by $E = \frac{F}{Q}$

Representation by electric field lines.

Magnitude of E in a radial field given by

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

Magnitude of E in a uniform field given by

$$E = \frac{V}{d}$$

- **Electric potential**

Understanding of definition of absolute electric potential, including zero value at infinity, and of electric potential difference.

Work done in moving charge Q given by

$$\Delta W = Q \Delta V$$

Magnitude of V in a radial field given by

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

Graphical representations of variations of E and V with r .

- **Comparison of electric and gravitational fields**

Similarities; inverse square law fields having many characteristics in common.

Differences; masses always attract but charges may attract or repel.